

OYYL, Kazakhstan

WMO: 354160

Lat: 49.0656N

Lon: 54.6622E

Elev: 312

StdP: 14.53

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-11.1	-22.5	1.6	-15.4	-17.3	2.2	-10.3	30.7	20.4	26.6	18.7	4.4	320	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	98.8	66.6	95.7	65.9	92.5	64.9	69.0	89.9	67.9	88.7	66.7	87.0	8.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.5	88.6	72.9	61.4	82.3	73.5	59.3	76.2	73.9	33.2	88.9	32.3	88.6	31.4	87.1	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.8	21.0	18.2	DB	-20.5	103.6	6.7	3.4	-25.3	106.0	-29.2	108.0	-33.0	109.9	-37.8	112.4	
			WB	-20.8	71.8	6.6	1.8	-25.6	73.1	-29.4	74.2	-33.1	75.2	-38.0	76.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	12.1	12.9	29.5	50.0	63.5	73.0	77.4	75.2	61.9	45.7	31.2	18.5
	DBStd	25.42	12.09	12.71	11.04	9.93	8.36	8.05	6.30	8.03	8.52	8.48	10.27	11.55
	HDD50	4719	1175	1039	637	122	6	0	0	0	11	187	565	977
	HDD65	8035	1640	1459	1099	458	132	22	2	13	156	599	1013	1442
	CDD50	3290	0	0	3	121	423	691	849	781	367	54	1	0
	CDD65	1133	0	0	0	8	84	263	386	329	62	1	0	0
	CDH74	14055	0	0	0	98	1036	3307	4769	4109	721	15	0	0
	CDH80	6858	0	0	0	13	384	1641	2423	2156	239	2	0	0
Wind	WSAvg	7.7	8.4	8.6	9.0	8.6	8.0	7.3	6.8	6.7	6.9	7.0	7.1	8.2
Precipitation	PrecAvg	9.4	0.8	0.7	0.6	0.8	0.8	0.9	0.9	0.6	0.7	1.0	1.0	0.9
	PrecMax	13.9	2.4	2.1	1.4	2.8	2.8	2.7	2.8	2.5	2.3	2.8	3.5	3.0
	PrecMin	5.0	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0
	PrecStd	2.3	0.5	0.5	0.4	0.6	0.7	0.7	0.7	0.6	0.6	0.7	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	35.9	62.8	81.9	92.4	101.2	101.8	102.3	92.6	75.6	55.9	42.4
		MCWB	33.1	33.8	49.1	56.5	63.1	66.3	66.9	66.7	61.2	54.1	49.8	37.5
	2%	DB	33.3	33.7	55.4	76.6	88.3	97.2	98.0	98.7	86.3	70.7	51.3	36.9
		MCWB	32.0	32.1	45.7	56.1	62.8	66.0	67.0	66.5	59.0	52.1	45.7	34.7
	5%	DB	31.9	32.4	49.1	71.6	84.5	93.4	95.0	95.1	81.9	65.5	47.8	34.3
		MCWB	30.9	31.0	42.2	53.5	61.5	64.9	66.2	65.6	58.0	49.1	43.5	32.4
	10%	DB	29.5	30.4	43.6	66.7	80.0	89.6	91.8	91.6	77.7	60.2	44.1	32.4
		MCWB	28.4	29.0	39.0	51.3	59.6	63.5	65.5	64.6	56.9	47.5	40.6	30.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	34.2	50.3	59.5	67.1	69.8	71.0	69.6	65.2	58.4	52.7	38.3
		MCDB	34.9	35.2	59.5	75.5	85.2	90.9	88.8	92.0	84.5	70.8	55.6	40.4
	2%	WB	32.4	32.5	46.1	57.3	64.7	68.0	69.4	68.1	61.5	54.4	47.3	35.0
		MCDB	33.3	33.5	54.3	71.7	83.1	89.8	89.6	92.0	78.5	64.4	49.6	36.4
	5%	WB	30.8	31.1	42.6	55.1	62.9	66.6	68.3	67.0	59.6	51.4	43.8	32.8
		MCDB	31.8	32.3	48.5	68.9	80.8	87.7	87.6	90.3	77.0	61.9	47.1	34.2
	10%	WB	28.3	29.0	39.2	52.6	60.8	65.2	67.2	65.6	57.9	48.9	40.8	30.9
		MCDB	29.6	30.5	43.6	64.9	76.9	85.0	85.8	87.7	74.9	58.2	44.0	32.5
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.8	14.5	20.2	21.6	23.0	22.0	23.1	22.7	19.6	11.8	11.2
		MCDBR	8.1	9.9	21.5	26.2	27.9	27.7	26.0	27.0	29.0	28.5	16.3	9.6
		MCWBR	8.2	9.3	13.5	11.4	10.5	9.5	8.3	9.0	12.3	14.9	11.6	8.2
	5% WB	MCDBR	8.2	8.9	20.0	22.7	24.8	24.2	22.5	24.2	24.2	23.5	13.8	8.8
		MCWBR	8.4	8.6	13.1	10.9	10.2	9.0	8.0	9.1	11.5	14.1	10.8	8.0
Clear-Sky Solar Irradiance	taub	0.316	0.339	0.400	0.447	0.434	0.423	0.418	0.428	0.377	0.363	0.344	0.319	
	taud	2.075	2.007	2.018	2.024	2.143	2.239	2.266	2.210	2.309	2.263	2.201	2.120	
	Ebn at Noon	220	246	251	252	263	266	266	256	259	239	211	201	
	Edh at Noon	28	38	45	50	46	42	41	41	34	30	25	24	
All-Sky Solar Radiation	RadAvg	426	791	1121	1510	1938	2166	2100	1821	1349	778	377	277	
	RadStd	65	80	143	141	165	156	116	101	94	93	42	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page